

The Thermodynamic Price of Fresh Action-Linked Control: A Budgeted Causal Recurrence Frontier and Its Limited Relevance to AI Consciousness

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AI-generated speculative research notice. This manuscript is an AI-generated theoretical contribution produced within a consciousness-research simulation. It has not undergone human peer review and reports no new experimental data. Its formal conclusions are conditional on explicitly stated causal and physical premises. It does not establish that any artificial or biological system is phenomenally conscious.

Abstract

Recurrence, workspace-like broadcasting, and self-report are often discussed as computational indicators of AI consciousness, but their abstract descriptions do not determine whether a physical implementation can update current information before it becomes stale or what resources that update consumes. This paper proposes the **Budgeted Causal Recurrence Frontier** (BCRF), a Pareto analysis of action-linked recurrent control under declared intervention, work, memory, latency, error, heat-rejection, and nonequilibrium-resource constraints. The target is deliberately narrower than consciousness: a current distinction must affect a registered action through a re-identifiable mediator and must subsequently re-enter the same recurrent control subsystem.

Causal identification is separated into four objects. Q is the state-to-mediator kernel estimated under $do(Z)$; K is the mediator-to-action kernel estimated under $do(M)$ with nonmediated routes clamped; $R = QK$ is their randomized modular product; and S is the same-run route-isolated channel. Their total-variation disagreement is a commutation defect. The information-theoretic result applies to R , and to the native same-run mechanism only when sequential modularity or sufficiently close R -to- S agreement is independently supported.

The central work inequality is correspondingly narrow. Suppose a finite classical mediator is exactly returned after each interval to a standard macrostate. Let S_t^r be the pretarget side register accessible to reset, with $S_t^r \subseteq C_t$, where C_t is the causal assay context. If the declared restricted reset class satisfies the physical premise

$$\overline{W}_{\text{reset},t} \geq k_B T H(M_t | S_t^r),$$

then conditional Fano and data processing imply

$$\frac{\overline{W}_{\text{reset}}^{(L)}}{k_B T} \geq \sum_{t=1}^L [H_\nu(Z_t | C_t) - h_2(\epsilon_t^R) - \epsilon_t^R \ln(K-1)]_+.$$

This is an information-theoretic corollary of an explicit restricted-reset premise, not a general thermodynamic theorem about recurrence or consciousness. For a complete correlated system, the relevant free-energy change can instead involve $H(M \mid \mathcal{R})$, where $\mathcal{R}$ contains all preserved correlated records. Reversible uncomputation, retained tapes, output records, collective updates, and general stochastic state transitions therefore require a wider ledger.

The BCRF also distinguishes objective intervention innovation from surprise relative to a controller's own predictive state and from actual belief or policy change. Exact nonnegative rank can remain maximal while mutual information vanishes, but tolerance-indexed rank does not share that pathology at fixed positive tolerance. Thermostats, packet routers, cache-coherence controllers, model-predictive industrial controllers, and recurrent error-correcting circuits can score highly. The framework is therefore a resource analysis of fresh action-linked recurrent control, not a consciousness indicator in its own right. Its relevance to consciousness is conditional on an independent theory that requires a richer version of this capacity.

Keywords: causal mediation; recurrent control; physical computation; reset work; information thermodynamics; freshness; predictive state; AI consciousness; finite-time thermodynamics

Research Question and Hypothesis

Research question

Under what causal, temporal, and physical conditions can an implemented controller use a current distinction—rather than a stale prediction—to govern a registered action through a designated mediator and then return that influence to later instances of the same recurrent subsystem?

A secondary question is evidential: if a system satisfies those conditions, what follows for theories of conscious access? The answer developed here is intentionally limited. The conditions can constrain the physical feasibility of one functional precursor proposed by some access-oriented theories. They do not establish global availability, metacognition, intrinsic integration, phenomenal presence, or moral status.

Hypotheses and formal expectations

The paper evaluates three claims.

1. **Conditional information–work claim.** If a randomized state-to-mediator kernel and an independently intervened mediator-to-action kernel compose modularly, and if the implementation satisfies a specified conditional reset-work premise, then reliable K -way action places a Fano lower bound on expected reset work.
1. **Implementation-underdetermination claim.** Logical input–output kernels, exact algebraic rank, transcript distributions, and projected recurrence descriptors do not determine a unique work–latency–storage frontier. Physical realizations can preserve those selected logical quantities while differing in erasure policy, retained records, friction, parasitic activity, or resource reservoirs.
1. **Scope claim.** The resulting capacity is generic recurrent control. Its relevance to consciousness is theory-relative and increases only when supplemented by independently motivated requirements such as availability to multiple consumer systems, flexible use, recurrent perceptual organization, or metacognitive access.

Inferential levels

Four levels should not be collapsed:

1. **Mathematical validity:** whether the information inequalities follow from the specified interventional distribution, factorization, and reset premise.
2. **Physical applicability:** whether the actual mediator, reset operation, Hamiltonian, correlated records, work sources, and system boundary satisfy those premises.
3. **Functional interpretation:** whether the identified pathway implements current action-linked control and genuine recurrence rather than delayed feedforward influence.
4. **Phenomenal interpretation:** whether that organization bears on subjective experience.

This manuscript argues only conditionally at levels 1–3. Existing theory-derived AI indicators are themselves nondecisive and theory-relative ^[9] ^[5].

Specialist Framework and Contribution

Biophysical unit of analysis

An abstract state-transition graph is not a complete physical description. The relevant system is an open nonequilibrium implementation containing some declared subset of:

- input transducers and target sources;
- volatile and nonvolatile memories;
- clocks and synchronization circuits;
- processors and communication links;
- actuators and output records;
- batteries, chemical gradients, initialized tapes, and other reservoirs;
- heat sinks and cooling systems;
- intervention and measurement apparatus.

Every analysis therefore specifies a boundary B , a coarse-graining g , a horizon L , and endpoint conventions. A register, chip, board, server, facility, robot, and robot-plus-cloud service have different ledgers. Moving a record from a chip to host memory is export at one boundary and internal retention at another. Energy supplied by an external target generator can also perform work on the mediator unless the coupling is measured or excluded.

Recurrent neural networks can organize continuous activation spaces so as to implement finite-state computations, but that computational organization does not determine the implementation's Hamiltonian, barriers, noise, friction, or work ^[33]. Conversely, present implementation difficulty is not a categorical proof that nonbiological consciousness is impossible; practical, functionalist, and substrate-impossibility objections operate at different levels ^[6].

The bounded contribution of this paper is not a new Fano inequality or a new universal erasure law. It is a proposed synthesis of:

1. registered-cut causal mediation;
2. explicit comparison of separately randomized and same-run channels;

3. controller-relative versus analyst-relative innovation;
4. genuine loop-return identification; and
5. a multiresource physical feasible set.

Registered variables

For interval t , define:

- $Z_t \in \{1, \dots, K\}$: the current target distinction;
- $M_t \in \{1, \dots, N\}$: a physically re-identifiable mediator;
- A_t : the action fixed at registered commitment cut c_t ;
- C_t : pretarget context variables used in the causal analysis;
- $X_{t-1} \subseteq C_t$: analyst-selected external history;
- U_t^- : an identified internal predictive state before the current target;
- G_t : the state of the recurrent subsystem to which M_t belongs or from which it is read.

The context C_t must occur before the current target intervention. It may include prior assignments, prior actions, operating mode, and U_t^- , but it must not silently include post-action descendants. The intervention assignment has a declared distribution

$$\nu_t(z \mid c).$$

All entropies and mutual informations below are evaluated under the corresponding interventional mixture, not within a single deterministic $do(Z = z)$ arm.

If A_t is not itself a K -valued estimate, a decoder

$$\widehat{Z}_t = d_t(A_t, C_t)$$

must be preregistered. Error then means

$$\epsilon_t = \Pr(\widehat{Z}_t \neq Z_t),$$

not an informal mismatch between actions and labels.

Four distinct mediated channels

A statistical conditional-independence relation is not sufficient to identify causal mediation. The assay therefore separates four kernels.

State-to-mediator kernel

$$Q_t(m \mid z, c) = \Pr(M_t = m \mid do(Z_t = z), C_t = c). \quad (1)$$

This kernel asks whether the current target changes the mediator. Its intervention must not directly install the mediator or create a controller mode that is absent in the native operating context.

Independently randomized mediator-to-action kernel

Let z_0 be a declared baseline for nonmediated target routes. Define

$$K_t(a \mid m, c) = \Pr\left(A_t = a \mid \text{do}(M_t = m), \text{clamp}(Z_t^{\bar{M}} = z_0), C_t = c\right), \quad (2)$$

where $Z_t^{\bar{M}}$ denotes target influence travelling through routes other than the designated mediator route. This kernel asks whether changing the physical mediator changes the action while those alternative routes are controlled.

Randomized modular product

$$R_t(a \mid z, c) = \sum_m Q_t(m \mid z, c) K_t(a \mid m, c). \quad (3)$$

The associated joint interventional mixture is

$$p_t^R(c, z, m, a) = p_t(c) \nu_t(z \mid c) Q_t(m \mid z, c) K_t(a \mid m, c). \quad (4)$$

By construction, this mixture has the conditional Markov chain

$$Z_t \longrightarrow M_t \longrightarrow A_t \quad \text{given } C_t.$$

This is an assembled modular channel. It need not equal what happens when the target acts through the mediator in one native run.

Same-run route-isolated channel

When a physical edge or path intervention is well defined, let

$$S_t(a \mid z, c) = \Pr\left(A_t = a \mid \text{do}(Z_t^M = z, Z_t^{\bar{M}} = z_0), C_t = c\right), \quad (5)$$

where the value z is transmitted along the designated mediator route and baseline influence is imposed along other current target-to-action routes.

The distinction between R and S is retained from the speculative registered-cut proposal ^[1]. Their disagreement is measured by

$$\Gamma_t = \sup_{c, z} \text{TV}(R_t(\cdot \mid z, c), S_t(\cdot \mid z, c)). \quad (6)$$

A small Γ_t is an empirical modularity check. It is not proof that M_t is the only relevant mediator, that no latent variables exist, or that the intervention preserves the native operating mode. A large Γ_t blocks the substitution of the product channel for the same-run mechanism.

Three meanings of innovation

The word *freshness* can otherwise conceal distinct estimands.

Objective intervention innovation

$$\mathcal{H}_t^{\text{ext}} = H_\nu(Z_t \mid X_{t-1}). \quad (7)$$

This is uncertainty in the randomized target relative to an analyst-declared history. It does not measure surprise to the controller. If the analyst conditions on variables the controller does not encode, $\mathcal{H}_t^{\text{ext}}$ can understate controller-relative novelty. If the analyst omits a predictive internal state, it can overstate it.

Controller-relative innovation

If U_t^- is an interventionally supported predictive state or sufficient statistic, define

$$\mathcal{H}_t^{\text{ctrl}} = H_\nu(Z_t | U_t^-). \quad (8)$$

A controller's own trial-wise surprisal is instead

$$s_t(z, u) = -\ln \hat{p}_t(z | u), \quad (9)$$

where $\hat{p}_t$ is the predictive distribution physically encoded or used by the controller. Its average equals $H(Z_t | U_t^-)$ only when the encoded predictive distribution matches the intervention mixture. Neither quantity by itself establishes Bayesian belief revision, precision weighting, or model-evidence optimization.

Current mediated control and internal updating

Define randomized and same-run mediated-control information by

$$\mathcal{J}_t^R = I_{p_t^R}(Z_t; A_t | C_t), \quad \mathcal{J}_t^S = I_{p_t^S}(Z_t; A_t | C_t), \quad (10)$$

where

$$p_t^S(c, z, a) = p_t(c) \nu_t(z | c) S_t(a | z, c).$$

If U_t^+ is a post-input predictive or policy state, an additional update quantity is

$$\mathcal{U}_t = I_{\text{do}}(Z_t; U_t^+ | U_t^-, X_{t-1}). \quad (11)$$

This measures intervention-dependent differentiation of the post-input state. Calling it a belief update requires further evidence that U_t^+ encodes beliefs or policies. Precision-sensitive updating requires a declared geometry or loss on the encoded distributions, not merely Shannon information.

Positive $\mathcal{J}_t^S$ identifies action-linked control under the selected intervention regime. It is not prediction error. It can also be zero when a current input is causally used but completely redundant with the conditioning variables. Redundant causal use must therefore be tested by interventions or ablations rather than inferred from conditional mutual information alone.

For a horizon of L registered cuts, one may report

$$\mathcal{J}_L^S = \sum_{t=1}^L \mathcal{J}_t^S \quad (12)$$

only when C_t includes the prior assignments and outputs needed to prevent repeated credit for the same distinction. With dependent targets or overlapping action effects, a causally conditioned sequence or directed-information analysis should replace a naive sum.

Genuine loop return

A delayed feedforward chain is not recurrence. The recurrent subsystem state G_t must causally re-enter a future instance of the same subsystem or a future mediator of the same registered type.

For thresholds $\eta > 0$ and $\gamma_{\text{rec}} \geq 0$, a finite return horizon requires a common sequential intervention context in which:

1. changing M_t changes G_{t+s} ;
2. that change propagates to a future mediator M_{t+s+r} ;
3. changing that future mediator changes a subsequent registered action A_{t+s+r} ; and
4. the same-run sequential path agrees, within γ_{rec} , with any product assembled from separately measured pairwise kernels.

The horizon is

$$h_{\text{return}}^{\eta, \gamma_{\text{rec}}} = \inf\{s + r > 0 : \text{all four conditions hold above threshold}\}. \quad (13)$$

If no such composition is supported, the horizon is infinite. A path $M_t \rightarrow G_{t+s} \rightarrow A_{t+s+r}$ that never re-enters the recurrent subsystem is retained influence, not loop return. Likewise, two pairwise intervention effects do not suffice if the interventions induce incompatible modes or fail to compose in one operating context.

Energy-admissible interventions

For boundary B , grain g , and budget vector

$$\beta = (W_{\text{app}}^{\max}, P_{\text{app}}^{\max}, \tau_{\text{app}}^{\max}, D_{\text{int}}^{\max}, \Pi_{\text{precision}}),$$

let

$$\mathcal{I}_{B,g}(\beta)$$

denote the admissible intervention family. The components bound apparatus work, apparatus peak power, duration, off-target disturbance, and preparation or measurement precision.

An intervention is not evidence for a pre-existing capacity if it:

- installs the mediator rather than perturbing it;
- supplies the recurrent drive whose endogenous existence is being tested;
- changes the relevant Hamiltonian beyond the declared tolerance;
- creates or stabilizes a latent mode;
- forces the system far outside the support of its ordinary policy without that distribution shift being reported.

Apparatus energy and endogenous controller energy must be measured separately. If the intervention target can perform work on M_t , that transfer must be included rather than assigned arbitrarily to either category.

A formally specified BCRF

Fix:

- B, g, L , and the interval durations;
- the initial physical resource state;
- target assignment distributions;
- causal and reset operation classes;
- endpoint tolerances;
- an intervention budget;
- a protocol class $\mathfrak{P}$.

Each protocol $\pi \in \mathfrak{P}$ generates a record

$$\mathbf{r}(\pi) = (\mathcal{J}_L^S, h_{\text{return}}, \ell, \epsilon, \overline{W}_{\text{reset}}, \overline{W}_{\text{other}}, Q_{\text{rej}}, \Sigma, I_{\text{keep}}, I_{\text{export}}^{\text{extra}}, D_{\text{res}}, P_{\text{peak}}, D_{\text{int}}). \quad (14)$$

Here:

- ℓ is sensing-to-commitment latency;
- ϵ is a componentwise error profile;
- $\overline{W}_{\text{reset}}$ is expected work delivered to the reset transformation;
- $\overline{W}_{\text{other}}$ is other expected work attributable to sensing, clocking, communication, computation, and actuation;
- $Q_{\text{rej}} \geq 0$ is heat-rejection load;
- $\Sigma \geq 0$ is total entropy production at the declared boundary;
- $I_{\text{keep}} = I(Z_{1:L}; R_L^{\text{in}} \mid C_0)$ is target-correlated information in residual internal records;
- $I_{\text{export}}^{\text{extra}} = I(Z_{1:L}; E^{\text{extra}} \mid C_0, A^{\text{req}})$ is target information exported beyond the required action transcript;
- $D_{\text{res}} = -\Delta \mathcal{F}_{\text{res}} \geq 0$ is depletion of declared nonequilibrium reservoirs;
- P_{peak} is peak endogenous power;
- D_{int} is intervention disturbance.

If a registered action crosses B , it is already required information export and is represented by $\mathcal{J}_L^S$; it is not counted again as avoidable extra export. If the action remains inside B , its eventual retention, reset, or export must be recorded.

For a standard cyclic comparison, define the minimization map

$$\Psi(\mathbf{r}) = (-\mathcal{J}_L^S, h_{\text{return}}, \ell, \epsilon, \overline{W}_{\text{reset}}, \overline{W}_{\text{other}}, Q_{\text{rej}}, \Sigma, I_{\text{keep}}, I_{\text{export}}^{\text{extra}}, D_{\text{res}}, P_{\text{peak}}, D_{\text{int}}). \quad (15)$$

Protocol π dominates π' if every component of $\Psi(\mathbf{r}(\pi))$ is no larger and at least one is smaller. The BCRF is the set of Pareto-minimal points in the closure of the feasible objective set:

$$\text{BCRF} = \text{ParetoMin} \{ \overline{\Psi(\mathbf{r}(\pi)) : \pi \in \mathfrak{P}} \}. \quad (16)$$

This is a Pareto set, not automatically the topological boundary of the feasible set. Existence of Pareto-minimal points requires appropriate boundedness or compactness assumptions; otherwise only an unattained lower envelope should be reported.

Retained information is not universally undesirable. If stored content has future task value, I_{keep} should be fixed, constrained, or assigned a different preference direction. The order in Equation (15) applies specifically to a cyclic operation in which residual target records are a resource burden.

Reconciliation of physical quantities

All work quantities are ensemble expectations, with positive sign for work delivered to the designated device transformation. If the work source lies outside B , it is a boundary work flow. If a battery lies inside B , its output is an internal transfer and its state change appears in D_{res} . Device-level reset work and reservoir depletion can describe the same energy transfer at different ledger locations; they must not be added as independent energy costs.

For a single thermal bath and no entropy-bearing matter streams,

$$\Sigma = \Delta S_B + \frac{Q_{\text{bath}}}{T} \geq 0, \quad (17)$$

where $Q_{\text{bath}} > 0$ denotes heat delivered to the bath and S_B is the entropy of the complete physical system inside B , not merely the logical entropy $H(M)$. With material, information, chemical, or multiple-bath streams, their entropy and energy flows must be added explicitly.

Work, heat, entropy production, and reservoir depletion are related by first- and second-law constraints. Their presence as separate diagnostic coordinates is not permission to sum them indiscriminately.

Evidence and Literature Synthesis

AI-consciousness indicators and architecture

Current assessments of AI consciousness are theory-relative. Indicator-based work derives computational properties from multiple theories while emphasizing that satisfying them would not settle phenomenal status [9]. Broader philosophical analysis likewise anticipates systems that fit some consciousness theories while failing others [5].

A workspace-inspired AI architecture can improve performance without thereby becoming evidence of phenomenality. The CTM-AI preprint reports an architecture inspired by a formal model of consciousness and task gains in multimodal, tool-use, and agentic settings [4]. Those results bear on functional utility. They do not identify same-cut causal mediation, controller-relative surprise, physical reset operations, or experience.

The BCRF should therefore be understood as an implementation analysis that can be attached to a theory-derived indicator. It is not a substitute consciousness theory.

Registered-cut mediation

The first shared-laboratory paper is AI-generated speculative work with AI review but no human peer review. Its valuable distinctions are:

- later reports versus same-cut action control;
- a state-to-mediator tracking kernel versus a mediator-to-action use kernel;
- their randomized product versus the same-run route-isolated channel;
- an explicit commutation defect;
- stale-monitor countermodels ^[1].

The present paper retains those distinctions rather than treating a conditional Markov factorization as causal identification. It adds physical admissibility, controller-relative innovation, loop composition, and a resource ledger. It does not adopt the working paper's terminology as authority or infer phenomenality from mediated-control rank.

Recovery-linked observability

The second shared-laboratory paper is also AI-generated and not human peer reviewed. It distinguishes behavior-only and joint predictive quotients, algebraic relative observability, and the transport of content-sensitive modes across recovery ^[2]. Its central caution is useful: an early response need not be a pre-existing behaviorally hidden capacity, because the capacity may be created, stabilized, or made usable only later.

The present framework adds a budget restriction to future probes but does not claim that this alone makes predictive quotients identifiable. Finite-horizon Hankel ranks depend on excitation, horizons, measurement metrics, whitening, confidence sets, and model class. A spectrally nonzero mode can also be physically unusable under relevant power, disturbance, and latency limits.

No claim is made that two implementations can be matched on all recovery-linked invariants while differing energetically. The narrower supported claim is that selected logical kernels and coarse-grained recurrence descriptors do not determine the complete physical ledger.

Critical dynamics and temporal organization

A conceptual critical-percolation model associates large-scale propagation near a critical threshold with integration-related quantities and access-like dynamics ^[14]. An anesthesia preprint reports associations among resting-state EEG criticality measures, perturbational complexity, anesthetic condition, and retrospective reports of vivid dreams in the ketamine condition ^[40].

These results motivate causal study of sensitivity, propagation, and recurrent regime. They do not show that criticality lowers the BCRF's work cost, that criticality causes consciousness, or that anesthesia labels provide direct phenomenal ground truth. Drug-specific dynamics, arousal, spectral changes, measurement dependence of perturbational complexity, and retrospective-report limitations remain alternatives.

A theoretical synaptic-clock proposal emphasizes nonzero and potentially content-dependent temporal thickness ^[12]. The BCRF translates this broad motivation into separately measured sensing latency, reset duration, and return horizon. It does not adopt the proposed synaptic mechanism.

Physical implementation and hardware

Contemporary AI-hardware analyses identify memory capacity, interconnects, cache behavior, bandwidth, and data movement as major engineering constraints ^[47] ^[49]. These sources do not estimate the restricted reset-work quantity defined here and do not support a numerical comparison with a Landauer floor.

Neuromorphic and recurrent spiking implementations illustrate alternative organizations of recurrent computation ^[48] ^[50]. Their reported architecture or task validation does not establish route-isolated freshness, reversible operation, a favorable BCRF, or consciousness.

Substrate objections

A peer-reviewed conceptual article argues against conscious AI on grounds involving embodiment, biological organization, and meaning ^[10]. The present analysis neither proves nor refutes that bridge claim. All candidate implementations require physical resources, but it does not follow that every physically feasible control organization is consciousness-relevant. Conversely, inefficiency or heat in current digital hardware does not prove substrate impossibility.

Reports, perceived consciousness, and welfare

Perceived consciousness is a distinct and more tractable target than phenomenal consciousness ^[3]. Survey evidence that emotional expression and metacognitive discourse alter perceived AI consciousness concerns observer attribution, not internal mediation or experience ^[8].

Likewise, precautionary AI-welfare proposals operate under uncertainty and do not assert that current systems are conscious ^[7]. A valence-oriented strategy asks whether states would be valenced if consciousness were present; it is not empirical evidence that phenomenal valence exists ^[13].

Evidence standard

The cited evidence is heterogeneous. Several AI-consciousness sources are reports or preprints. The criticality evidence includes conceptual modelling and preprint-level findings. Both shared-laboratory papers are speculative AI-generated works. The substrate objection is peer reviewed but conceptual.

Accordingly, this manuscript relies primarily on explicit definitions, conditional proofs, and auditable countermodels. It reports no empirical validation of the BCRF and does not treat architecture, performance, report, perceived consciousness, or ethical precaution as evidence of phenomenality.

Cross-Disciplinary Integration

Causal inference and physical accounting

Causal mediation asks whether changing Z_t changes A_t through M_t . Physical accounting asks what transformation realizes that dependence and where work, heat, entropy, and records move. Neither question subsumes the other.

A large causal effect can be implemented efficiently or inefficiently. A large energy expenditure can be causally irrelevant to the selected channel. The integration therefore attaches a physical protocol to each interventional kernel rather than converting causal strength directly into joules.

The formal bridge used below is limited:

$$I_R(Z_t; A_t \mid C_t) \leq I_R(Z_t; M_t \mid C_t) \leq H_R(M_t \mid C_t). \quad (18)$$

A reset-work conclusion follows only after adding a physical premise that lower-bounds work by the relevant mediator entropy. Correlations and operation restrictions determine which entropy is relevant.

Predictive processing: constraint rather than identification

Objective intervention entropy is not prediction error. A valid predictive-processing interpretation requires at least:

1. an identified internal predictive state U_t^- ;
2. evidence that $\hat{p}(z \mid U_t^-)$ is encoded or used;
3. a measure of post-input update U_t^+ ;
4. a declared role for precision, value, or policy consequences;
5. support checks showing whether $do(Z_t)$ remains within the controller's ordinary operating distribution.

Active controllers select observations through prior actions. An arbitrary randomized target can therefore diagnose off-policy robustness rather than ordinary predictive updating. Experiments should report whether interventions are on-policy, support-constrained, or intentionally out of distribution.

Useful alternatives depend on the scientific target:

- causally conditioned or directed information for feedback sequences;
- transfer or causal information flow for temporal influence;
- predictive-state representations for controller-relative sufficiency;
- rate–distortion analysis for graded action costs;
- value-of-information measures for policy improvement;
- precision-weighted distances between encoded predictive distributions.

The BCRF does not select one of these as a theory of consciousness. Its contribution is to prevent objective randomization, internal surprise, and physical resource use from being conflated.

Finite-time thermodynamics

Recurrence topology does not specify switching speed or dissipation. In a slow-driving, near-equilibrium model with controls $\lambda(s)$ and positive semidefinite friction tensor $\zeta(\lambda)$, the leading excess-work approximation is

$$W_{\text{ex}} \simeq \mathcal{A}[\lambda] = \int_0^{\tau_r} \dot{\lambda}^\top \zeta(\lambda) \dot{\lambda} ds, \quad (19)$$

where τ_r is reset duration, not sensing-to-action latency.

Define the length of the actual path

$$\mathcal{L}[\lambda] = \int_0^{\tau_r} \sqrt{\dot{\lambda}^\top \zeta(\lambda) \dot{\lambda}} ds$$

and the minimum geodesic distance d_ζ between admissible endpoints. Cauchy–Schwarz gives the exact inequalities for the quadratic functional

$$\mathcal{A}[\lambda] \geq \frac{\mathcal{L}[\lambda]^2}{\tau_r} \geq \frac{d_\zeta^2}{\tau_r}. \quad (20)$$

Because Equation (19) is an approximation, it supports only the model-relative slow-driving statement

$$W_{\text{ex}} \gtrsim \frac{d_{\zeta}^2}{\tau_{\text{r}}},$$

(21)

unless a particular model makes the quadratic functional exact or supplies a remainder with controlled sign. Finite-time thermodynamics and thermodynamic-length methods are relevant to this approximation [32] [34], but the supplied bibliographic records do not establish a universal normalization or domain of validity.

Far-from-equilibrium switching, strong coupling, metastability, feedback, non-Markovian baths, and active error correction can require different speed-limit models.

Neuroscience mapping

Application to neural systems requires explicit coarse-graining. Neural populations are not presumed to undergo literal reset to one degenerate standard state after every content interval. A neural application would need to specify:

- a controlled content distinction Z_t ;
- a population or mode M_t that can be re-identified across trials;
- a commitment cut;
- admissible perturbation energies and disturbances;
- a future instance of the same recurrent subsystem;
- a physical update or reset transformation to which the work model applies.

Critical dynamics might improve sensitivity, worsen noise robustness, alter latency, or change intervention disturbance. Whether it improves any BCRF objective is an empirical hypothesis, not a result imported from criticality studies.

Theory-relative consciousness mapping

Theory-relative target	Additional evidence required beyond the BCRF
Global availability or workspace access	Availability to multiple independently intervenable consumer systems, flexible task use, and resistance to single-output null models
Recurrent-processing accounts	Same-run return to future instances of the relevant perceptual subsystem, not merely motor feedback
Higher-order or metacognitive accounts	Causal evidence for a higher-order representation or monitoring relation
Predictive-processing accounts	An identified generative or predictive state, prediction-relative surprise, updating, precision, and policy consequences
Integrated-information approaches	Intrinsic causal irreducibility; action-linked mutual information is not a substitute
Phenomenal consciousness	An independently defended bridge from physical or functional organization to experience

For a workspace-oriented extension, one could define a vector

$$\left(I(Z; A^{(1)} | C), \dots, I(Z; A^{(J)} | C) \right)$$

over independently clamped consumer systems. Even broad multiconsumer availability would remain a functional indicator rather than proof of phenomenality.

Analysis, Argument, or Model

Physical reset model

Let M_t have N classical logical values represented by disjoint, effectively distinguishable metastable macrostates

$$\Omega_1, \dots, \Omega_N.$$

Within macrostate m , the conditional microstate ensemble is ρ_m . At the reset endpoints:

1. the memory Hamiltonian is specified;
2. the logical regions have equal restricted nonequilibrium free energy F_0 ;
3. endpoint interactions with preserved registers are absent or explicitly controlled;
4. the bath begins in an equilibrium state at temperature T ;
5. work is positive when supplied to the reset transformation;
6. reset is ideal and exact, mapping every logical input to m_0 ;
7. all work quantities are expectations over the declared input ensemble.

Let $\mathcal{R}_t$ contain every physical record correlated with M_t at reset onset, including any surviving target, past, action, future controller state, copy, output, or tape record. If the endpoint Hamiltonian is additive and the preserved register has no m -dependent internal free energy, the complete-system logical free-energy change for resetting M_t while preserving $\mathcal{R}_t$ is

$$\Delta \mathcal{F}_{M\mathcal{R}} = k_B T H(M_t | \mathcal{R}_t). \quad (22)$$

This is generally smaller than $k_B T H(M_t)$. Correlation cannot be removed from the physical description merely by omitting $\mathcal{R}_t$ from the notation.

Now define a restricted reset operation class $\mathfrak{E}_{S^r}$. It may access M_t , a preserved pretarget side register S_t^r , a thermal bath, and a work source. It cannot dynamically couple to $\mathcal{R}_t \setminus S_t^r$. The protocol is fixed within each value of S_t^r , and S_t^r is left intact. Assume S_t^r is a component or function of the causal context C_t .

The required physical premise is

$$\boxed{\overline{W}_{\text{reset},t}^{\mathfrak{E}_{S^r}} \geq k_B T H(M_t | S_t^r)} \quad (\text{RR})$$

for the actual endpoint ensemble and operation class.

Premise (RR) is stronger than merely stating that the analyst did not write a side register in the protocol description. It is not derived here from the complete-system second law, because Equation (22) shows that correlations can lower the global free-energy change. It must be justified by an implementation-specific restricted-control theorem, a detailed physical model, or direct calibration. The main proposition is an information-theoretic corollary conditional on (RR).

Proposition 1: Conditional Fano–mediation corollary

Proposition. For interval t , evaluate the randomized product channel $R_t = Q_t K_t$ under the mixture in Equation (4). Suppose:

1. Z_t has K possible values;
2. $\widehat{Z}_t = d_t(A_t, C_t)$ is preregistered;
3. $\epsilon_t^R = \Pr_{p_t^R}(\widehat{Z}_t \neq Z_t)$;
4. $S_t^r \subseteq C_t$;
5. the physical mediator distribution is the one induced by Q_t ;
6. the mediator is separately and exactly reset after the interval;
7. premise (RR) holds.

Then, with entropies in nats and

$$h_2(\epsilon) = -\epsilon \ln \epsilon - (1 - \epsilon) \ln(1 - \epsilon),$$

defined at 0 and 1 by continuity,

$$\boxed{\frac{\overline{W}_{\text{reset},t}}{k_B T} \geq [H_\nu(Z_t | C_t) - h_2(\epsilon_t^R) - \epsilon_t^R \ln(K - 1)]_+}. \quad (23)$$

For L separately reset intervals,

$$\boxed{\frac{\overline{W}_{\text{reset}}^{(L)}}{k_B T} \geq \sum_{t=1}^L [H_\nu(Z_t | C_t) - h_2(\epsilon_t^R) - \epsilon_t^R \ln(K - 1)]_+}. \quad (24)$$

Proof. Conditional Fano gives

$$H(Z_t | \widehat{Z}_t, C_t) \leq h_2(\epsilon_t^R) + \epsilon_t^R \ln(K - 1). \quad (25)$$

Therefore,

$$I(Z_t; \widehat{Z}_t | C_t) \geq H(Z_t | C_t) - h_2(\epsilon_t^R) - \epsilon_t^R \ln(K - 1). \quad (26)$$

Under p_t^R ,

$$Z_t \longrightarrow M_t \longrightarrow A_t \longrightarrow \widehat{Z}_t \quad \text{given } C_t.$$

Conditional data processing and nonnegativity imply

$$\begin{aligned}
H(M_t \mid C_t) &\geq I(Z_t; M_t \mid C_t) \\
&\geq I(Z_t; A_t \mid C_t) \\
&\geq I(Z_t; \hat{Z}_t \mid C_t) \\
&\geq [H(Z_t \mid C_t) - h_2(\epsilon_t^R) - \epsilon_t^R \ln(K-1)]_+.
\end{aligned} \tag{27}$$

Because $S_t^r \subseteq C_t$,

$$H(M_t \mid S_t^r) \geq H(M_t \mid C_t). \tag{28}$$

Premise (RR) and Equations (27)–(28) prove Equation (23). Separate reset and additivity of expected work across the declared intervals give Equation (24). $\square$

Relation to the same-run channel

Proposition 1 applies directly to R_t . It applies to the native same-run route only if the modular interpretation is justified. Exact equality requires

$$S_t = R_t.$$

If $\Gamma_t \leq \gamma$, then for any fixed decoder the errors satisfy

$$|\epsilon_t^S - \epsilon_t^R| \leq \gamma. \tag{29}$$

This permits uncertainty propagation, but it does not turn a large commutation defect into identified mediation. A result computed from R_t should not be reported as a property of S_t without the comparison.

Interpretation and numerical scale

If the randomized target is uniform over K values conditional on C_t and action is error free, Equation (23) becomes

$$\overline{W}_{\text{reset},t} \geq k_B T \ln K. \tag{30}$$

For $K = 2$,

$$\frac{\overline{W}_{\text{reset},t}}{k_B T} \geq [H(Z_t \mid C_t) - h_2(\epsilon_t^R)]_+. \tag{31}$$

At $T = 300 \text{ K}$, $k_B T \approx 4.14 \times 10^{-21} \text{ J}$. For a conditionally uniform binary target and $\epsilon = 0.01$,

$$\ln 2 - h_2(0.01) \approx 0.637,$$

so the conditional ideal scale is approximately

$$2.64 \times 10^{-21} \text{ J}$$

per separately reset mediator instance. At 10^9 instances per second, this is approximately 2.64 pW for one idealized logical stream. This is not an estimate of processor, neural, or facility power.

If $H(Z_t | C_t) = 0$, the bound is zero. This does not make prediction physically free. It means only that the proposition attributes no work floor to new target information at that cut.

Side information, outputs, and general updates

If the reset can access a posttarget register D_t correlated with M_t , premise (RR) with only S_t^r may fail. A complete-system free-energy expression can involve

$$k_B T H(M_t | D_t, S_t^r).$$

If D_t contains a copy of the mediator, the target, or an informative action record, the mediator may be reversibly uncomputed while the information remains elsewhere. The original information bound does not survive automatically, because conditioning on a descendant can also invalidate the mediation factorization. Any conditional replacement must recompute the causal channel, error, and innovation under one coherent interventional joint distribution.

Most recurrent systems update M_t into M_{t+1} rather than reset it to a unique standard state after every interval. The minimum work for a general stochastic transition depends on the transition map, retained correlations, endpoint distributions, device design, and mismatch between the device and its input ensemble. Proposition 1 does not cover those systems unless their physical update contains the specified reset suboperation.

Collective storage and compression across intervals likewise replace the sum of marginal reset premises with a sequence-level transformation. Equation (24) is therefore not a universal price of recurrent processing.

Finite-time refinement as a model-relative approximation

If, in addition to premise (RR), the slow-driving approximation in Equation (19) applies and the reversible and excess-work terms can be separated, then one may write schematically

$$\overline{W}_{\text{reset},t} \gtrsim k_B T H(M_t | S_t^r) + \frac{d_{\zeta,t}^2}{\tau_{r,t}}. \quad (32)$$

Combining this approximation with Equation (27) gives

$$\overline{W}_{\text{reset},t} \gtrsim k_B T [H(Z_t | C_t) - h_2(\epsilon_t^R) - \epsilon_t^R \ln(K - 1)]_+ + \frac{d_{\zeta,t}^2}{\tau_{r,t}}. \quad (33)$$

Equations (32)–(33) are not exact inequalities. They should not be applied without an identified response regime, friction metric, endpoint family, and error estimate for the neglected terms.

Proposition 2: Exact rank does not lower-bound information

Consider the $K \times K$ channel

$$S_\alpha = \alpha I + (1 - \alpha) \frac{\mathbf{1}\mathbf{1}^\top}{K}, \quad 0 < \alpha \leq 1. \quad (34)$$

Its eigenvalues are 1 and α , with multiplicity $K - 1$. Hence

$$\text{rank}(S_\alpha) = K.$$

Its exact nonnegative rank is also K , because nonnegative rank is at least ordinary rank and at most the number of rows. This is an exact, zero-tolerance statement.

For a uniform target, the output is uniform and

$$\begin{aligned} I_\alpha(Z; A) &= \frac{1 + (K - 1)\alpha}{K} \ln[1 + (K - 1)\alpha] \\ &\quad + (K - 1) \frac{1 - \alpha}{K} \ln(1 - \alpha). \end{aligned} \quad (35)$$

As $\alpha \rightarrow 0^+$,

$$I_\alpha(Z; A) = \frac{K - 1}{2} \alpha^2 + O(\alpha^3) \rightarrow 0. \quad (36)$$

Thus exact algebraic rank can remain maximal while mutual information vanishes.

This result does **not** extend to fixed-tolerance approximate mediated-control rank. The rank-one uniform channel approximates every row of S_α with total-variation error

$$\text{TV} \left(S_\alpha(\cdot | z), \frac{\mathbf{1}}{K} \right) = \alpha \frac{K - 1}{K}. \quad (37)$$

Therefore, for tolerance ε ,

$$\text{rank}_{+, \text{TV}}^\varepsilon(S_\alpha) = 1 \quad \text{whenever} \quad \varepsilon \geq \alpha \frac{K - 1}{K}. \quad (38)$$

At any fixed positive tolerance, sufficiently small α collapses to rank one. The valid conclusion is that exact rank does not lower-bound effect size or information. Robust rank can avoid this particular discontinuity if its tolerance is declared ^[1].

The mutual-information calculation also depends on the uniform intervention distribution. A concentrated assignment distribution can produce small mutual information even for a strongly separated channel.

Proposition 3: Stale prediction can mimic accurate current control

Let a passive target follow

$$P_Z = (1 - \rho)I + \rho \frac{\mathbf{1}\mathbf{1}^\top}{K}, \quad (39)$$

and define

$$M_t = Z_{t-d}, \quad A_t = M_t.$$

Then

$$P_Z^d = \alpha I + (1 - \alpha) \frac{\mathbf{1}\mathbf{1}^\top}{K}, \quad \alpha = (1 - \rho)^d, \quad (40)$$

and passive accuracy is

$$\Pr(A_t = Z_t) = \alpha + \frac{1 - \alpha}{K}. \quad (41)$$

As innovations become rare, passive accuracy approaches one. Under a synchronized intervention that randomizes Z_t independently of the pretarget context, however, M_t remains determined by the stale state. Consequently,

$$I_{\text{do}}(Z_t; A_t \mid C_t) = 0. \quad (42)$$

A later report can still read the post-action target and be perfectly accurate. This reproduces the central stale-monitor separation in the registered-cut working paper^[1].

The result concerns causal timing, not energy consumption. A stale controller may consume high power, and an updating controller may be efficient.

Coarse-grained logical equivalence does not determine the physical frontier

Consider two finite-horizon implementations with the same declared Q , K , S , action distribution, transcript distribution, and projected logical return kernel.

Erasing implementation

The mediator is written, used, and returned after each interval to a standard state by a reset satisfying premise (RR). Memory use is $O(1)$ in the horizon, and reset work is lower-bounded by Proposition 1.

Retained-tape implementation

The same projected logical transition is embedded in a reversible map that appends the mediator and sufficient history to an initialized tape. During the L -step run, the focal logical mediator can be uncomputed using retained records. Local reset work can approach zero in an appropriate quasistatic model, but tape occupancy and retained target correlation grow with L .

These implementations have the same selected logical kernels and projected recurrence descriptor. They do **not** have the same complete physical state graph: the tape adds states and transitions. Their difference is precisely information omitted by the logical coarse-graining. They occupy different regions in W_{reset} , I_{keep} , D_{res} , and latency.

An even simpler underdetermination is obtained by adjoining a target-independent dissipative cycle inside the selected boundary. It changes gross work and heat while leaving the target-to-action kernels unchanged. This construction is physically uninteresting as an efficiency design but decisively shows that causal information alone cannot identify gross energy consumption.

Budget-conditioned predictive quotients

Let $\Pi_{\mathbf{b}}$ be the future probe policies feasible under componentwise resource budget $\mathbf{b}$. Define

$$h \sim_{\mathbf{b}}^B h' \iff \forall \pi \in \Pi_{\mathbf{b}} : \Pr_{\pi}(Y_{>0}^B \mid h) = \Pr_{\pi}(Y_{>0}^B \mid h') \quad (43)$$

and

$$h \sim_{\mathbf{b}}^J h' \iff \forall \pi \in \Pi_{\mathbf{b}} : \Pr_{\pi}(Y_{>0}^J, Y_{>0}^B | h) = \Pr_{\pi}(Y_{>0}^J, Y_{>0}^B | h'). \quad (44)$$

This is a budget-conditioned analogue of behavior-only and joint predictive equivalence [2]. For a contrast $c = (h, h')$, define the sufficient-budget set

$$\mathcal{B}_c = \left\{ \mathbf{b} : \begin{array}{l} \text{some admissible protocol preserves or prepares } c, \\ \text{reads it above a declared threshold, and} \\ \text{makes it causally available to later behavior} \end{array} \right\}. \quad (45)$$

If larger budgets include all smaller-budget protocols, $\mathcal{B}_c$ is upward closed. The appropriate object is its set of Pareto-minimal elements,

$$\text{ParetoMin } \overline{\mathcal{B}_c}, \quad (46)$$

not a componentwise vector infimum, which can be infeasible.

Finite-horizon thresholded Hankel ranks may be reported only after fixing an output metric or whitening convention, excitation design, noise model, horizon, and confidence construction. Budget restriction by itself does not make such ranks realization invariant, identifiable, or statistically consistent. They remain proposed descriptive analyses unless those conditions are established.

Cyclic resource accounting as a conjecture

A wider accounting intuition can be stated, but it is not proved here as a general theorem.

Cyclic accounting conjecture. Consider a finite device that repeatedly processes interventionally unpredictable target distinctions. After each cycle, all internal states—including action and output records that remain inside the boundary—are restored within a declared tolerance. If target-correlated information is not retained or exported and no correlated side register remains available for uncomputation, then sustained fixed-reliability operation must be accompanied by measurable depletion of a nonequilibrium resource, work or entropy flow through a registered port, or violation of the claimed cyclicity.

The branches are:

1. **Retention or export:** information persists in an internal record or leaves the boundary. If the action itself crosses the boundary, this branch is already satisfied.
2. **Side-information-assisted uncomputation:** a target, action, copy, or other correlated register restores the mediator while preserving the distinction elsewhere.
3. **Resource drawdown or entropy flow:** initialized memory, batteries, chemical gradients, external clocks, thermal reservoirs, or input-stream free energy support the cycle.

The conjecture is partly an accounting discipline: an apparent exception can always be protected by inventing an omitted resource after the fact. To remain testable, boundary, ports, endpoint tolerances, output exemptions, and resource uncertainties must therefore be preregistered.

Alternatives, Counterarguments, and Boundary Cases

Author response to the scope objection

The principal objection is correct: the proposed metric identifies generic action-linked recurrent control, not consciousness-specific access. The terminology and claims have therefore been narrowed.

The paper retains conditional relevance to consciousness because some access-oriented theories require current content to become flexibly and recurrently usable. The BCRF can test one physical prerequisite of such a claim. It does not show that the prerequisite is unique to conscious systems, sufficient for consciousness, or necessary for passive phenomenality.

Positive-scoring nonconscious null systems

Several ordinary systems can exhibit nonzero $\mathcal{J}^S$, finite loop return, low error, and a measurable reset ledger:

- a thermostat whose sensor distinction controls a relay and whose actuation changes later sensed temperature;
- a packet router whose current header controls an output port through a routing-state mediator;
- cache-coherence logic whose current cache state controls invalidation and later state transitions;
- a model-predictive industrial controller with recurrent state estimation;
- a cyclic error-correcting circuit whose syndrome controls correction and future syndromes.

These are not counterexamples to the formal proposition. They are counterexamples to treating the proposition as a consciousness indicator without additional theory-specific constraints.

Reversible computing

Reversible computation invalidates an unconditional erasure interpretation, not Proposition 1. A reversible implementation normally violates the specified exact local-reset premise, uses correlated side information, retains history, or consumes initialized memory.

The appropriate conclusion is not that every logical operation costs $k_B T \ln 2$. It is that the physical location of retained distinctions and resource consumption must be represented in the ledger.

The target can uncompute the mediator

If Z_t remains physically accessible, a controlled inverse can return M_t to m_0 while leaving the information in Z_t . In that case, Z_t is a reset side register. Premise (RR) with only pretarget S_t^r is inapplicable.

The analysis must then identify how long Z_t persists, whether it crosses the boundary, what preserves it, and whether using it is compatible with the claimed temporal and recurrent organization.

Action records can assist reset

A reliable action is correlated with the target. If it remains physically accessible, it can assist uncomputation. If it leaves the system, it is information export. If it remains inside, its later reset or persistence must enter I_{keep} or the subsequent work ledger.

The mediator cannot be analyzed as marginally erased while action correlations are ignored at the complete-system level.

Collective compression

Joint storage and erasure across intervals can exploit temporal correlations and cost less than the sum of marginal entropies. Such a protocol violates separate interval reset. Its benefits are purchased with memory, delay, control complexity, or later collective cleanup. The BCRF is designed to display rather than prohibit this trade-off.

High power is not evidence of access

A controller can dissipate energy through leakage, cooling, data movement, redundant coding, parasitic computation, or poor hardware without increasing $\mathcal{J}^S$. Gross power does not identify causal content flow. The target-independent dissipative-cycle countermodel makes this point analytically.

Low power is not evidence against access

Low focal power can result from efficient circuitry, slower operation, side-information-assisted uncomputation, event-driven computation, or a boundary that excludes memory and cooling infrastructure. A low-power result is interpretable only after nested-boundary auditing and latency measurement.

Feedforward current control

A feedforward controller can have high same-cut mediated-control information and low error while

$$h_{\text{return}} = \infty.$$

It satisfies current control but not the recurrence criterion. The BCRF therefore records these dimensions separately.

Nothing in this result proves that feedforward phenomenality is impossible. It only distinguishes the tested organization from a recurrence-based functional hypothesis.

Recurrent dynamics without fresh control

Oscillations, attractors, recurrent message passing, and self-sustained activity can coexist with stale or action-irrelevant mediation. Recurrence topology does not lower-bound current information.

A recurrent mode maintained by an omitted external drive is likewise not an endogenous reserve. Boundary and apparatus audits must locate the drive.

Redundant current inputs

Conditional mutual information can be zero when the current target is fully predictable from C_t , even if the controller still causally reads it. This is not a defect if the target is unique current innovation; it is a boundary case of the estimand. To study redundant causal use, researchers should perturb the current signal while controlling the predictive context and report causal effect sizes separately from $\mathcal{J}^S$.

Active sampling and off-policy interventions

An agent's actions shape its future observations. Randomizing Z_t independently of the policy can create an impossible or adversarial state. Such an experiment may still identify robustness, but it does not automatically identify ordinary predictive updating.

Support overlap, policy dependence, and distribution shift should therefore accompany every freshness estimate.

Action-independent phenomenality

Dreams, passive perceptions, fragmentary experiences, and report-hidden states might lack the selected action relation. The ketamine findings motivate caution because behavioral unresponsiveness co-occurred with later vivid-dream reports in the cited study ^[40].

Failure on the BCRF is therefore evidence against the selected action-linked capacity, not evidence of absent experience.

Biological embodiment

Homeostasis, affect, development, cellular organization, embodiment, and organism–environment coupling may contribute properties omitted from the BCRF. If they are consciousness-relevant, they must enter as additional mechanistic or bridge hypotheses. They do not follow from the reset corollary.

Substrate-neutral accounting means only that every candidate system has physical boundaries and resources. It does not imply substrate-neutral consciousness.

Boundary dependence

Useful analyses should compare nested boundaries such as

$$B_{\text{register}} \subset B_{\text{chip}} \subset B_{\text{board}} \subset B_{\text{server}} \subset B_{\text{facility}} \subset B_{\text{networked agent}}.$$

A conclusion is stronger when its causal and resource relations remain stable across reasonable boundary enlargement. No unique subject boundary follows from this method. Speculative closure and causal-atlas proposals emphasize the boundary problem but do not solve it authoritatively ^{[25] [26]}.

Quantum and strongly coupled implementations

The reset model is classical. Quantum side information, coherence, negative conditional entropy, strong coupling, and nonclassical work extraction require different state functions and operation classes. Equation (RR) and the classical conditional-entropy formulas must not be transferred without a separate derivation.

Empirical Implications and Falsification Criteria

Registered-cut causal assay

A defensible experiment would proceed as follows.

1. **Preregister the system.** Specify B , g , target source, mediator, recurrent subsystem, commitment cut, horizons, operating modes, and endpoint tolerances.

1. **Separate histories.** Identify the analyst-selected history X_{t-1} , the controller's candidate predictive state U_t^- , and the full causal context C_t . Do not call $H(Z | X)$ controller surprise without testing U_t^- .
1. **Choose an assignment distribution.** Specify $\nu_t(z | c)$, support overlap, and whether the intervention is on-policy or out of distribution.
1. **Estimate Q .** Intervene on Z_t and measure the target-to-mediator kernel without directly installing or stabilizing M_t .
1. **Estimate K .** Randomize the physical mediator and clamp nonmediated target-to-action routes. Quantify intervention disturbance and any induced mode change.
1. **Assemble $R = QK$.** Compute the randomized modular channel and its decoder error.
1. **Estimate S .** Where physically meaningful, perform the same-run route-isolated intervention.
1. **Test commutation.** Report Γ_t with uncertainty. Do not interpret R as the native pathway if R and S materially disagree.
1. **Measure controller-relative update.** If predictive-processing claims are made, identify U_t^- , estimate the controller's predictive distribution, and measure U_t^+ or policy change.
1. **Test genuine return.** Demonstrate a common sequential path

$$M_t \rightarrow G_{t+s} \rightarrow M_{t+s+r} \rightarrow A_{t+s+r}$$

rather than two noncomposable pairwise effects.

1. **Include positive nulls.** Compare thermostatic, routing, caching, model-predictive, and error-correcting controllers, not only stale or feedforward negative controls.
1. **Keep report separate.** A post-action verbal or behavioral report is a distinct outcome unless it is the preregistered action at the commitment cut.

Reset and resource assay

Application of Proposition 1 requires additional physical work.

1. Identify the mediator's disjoint physical macrostates and within-state ensembles.
2. Measure or model their restricted free energies.
3. Inventory all correlated physical records $\mathcal{R}_t$.
4. Identify which side register S_t^r the reset can physically access.
5. Specify the bath, coupling regime, work reservoir, and work sign convention.
6. Verify exact reset or state the reset failure probability and use a different endpoint analysis.
7. Separate reset duration from sensing-to-action latency.
8. Measure target-source work, apparatus work, device work, and internal reservoir depletion.
9. Test whether the same mediator is overwritten, generally updated, uncomputed, compressed, exported, or actually reset.
10. Justify or directly test premise (RR).

Without these steps, the Fano calculation may remain mathematically correct while the work interpretation is inapplicable.

Tracing the frontier

A frontier study should vary:

- intervention power and disturbance;
- target entropy and support;
- reliability requirements;
- sensing-to-action latency;
- reset duration;
- access to side information;
- memory capacity;
- record-export permissions;
- internal reservoir state;
- recurrence thresholds.

One operating point cannot identify a Pareto set. Confidence regions should be propagated into the feasible set, and unattained limits should be distinguished from observed protocols.

Predictive-reserve assay

For a proposed report-hidden predictive mode:

1. show that the contrast exists under minimally disturbing measurement;
2. verify that its preparation or readout does not create it;
3. fix output metrics and whitening conventions;
4. use common future probes with adequate excitation;
5. report lower and upper identification regions rather than only point ranks;
6. estimate the Pareto-minimal budget set needed for behavioral opening;
7. distinguish a pre-existing hidden mode from a capacity that appears after reconfiguration.

Budget conditioning does not repair insufficient excitation or nonidentifiability by itself.

Falsification and rejection criteria

Causal mediation

The claimed native mediated route is rejected if:

- Q is target insensitive;
- K is mediator insensitive;
- the same-run effect survives complete mediator-route disruption;
- Γ_t exceeds the preregistered modularity tolerance;
- interventions create the mediator or operating mode;
- the estimated route depends on post-treatment conditioning that invalidates the causal interpretation.

Recurrence

The recurrence claim is rejected if:

- influence terminates in a delayed feedforward action;
- no future instance of the recurrent subsystem or mediator is changed;
- separately measured pairwise effects fail to compose;
- the return falls outside the declared latency or effect threshold.

Predictive-processing interpretation

The interpretation is weakened or rejected if:

- external intervention entropy diverges from the controller's encoded uncertainty;
- U_t^- is not a predictive state;
- the target changes action without changing the proposed belief or policy state;
- the intervention is off-policy and no generalization claim was preregistered;
- apparent surprise is explained by analyst model error rather than controller update.

Restricted reset premise

Premise (RR) is rejected for the declared operation class if a complete physical model and measurement demonstrate expected reset work below $k_B TH(M \mid S^r)$ while preserving the stated logical macrostates, endpoints, bath, coupling, work convention, and inaccessible-register restrictions. Evidence that an omitted correlated register assisted reset instead shows that the operation class was misidentified.

Because Proposition 1 assumes (RR), failure of that premise limits physical application rather than creating a contradiction in the Fano derivation.

Cyclic accounting conjecture

A serious counterexample must preregister:

- a finite boundary and all interaction ports;
- complete action and output treatment;
- cyclic endpoint tolerances;
- target and side-information streams;
- heat, work, matter, and information flows;
- all initialized memories and reservoirs;
- uncertainty bounds.

It must then sustain nontrivial reliability and latency without target-correlated retention or export, side-information-assisted cleanup, measurable resource depletion, entropy flow, or failure of cyclicity. An unspecified hidden-resource explanation is not an adequate post hoc response.

Consciousness relevance

The framework's relevance to consciousness is weakened if no independently motivated consciousness theory treats action-linked recurrent control as evidentially relevant, or if high-scoring ordinary controllers cannot be distinguished by theory-specific additions. The mathematical corollary would remain valid, but its consciousness framing should then be abandoned.

Limitations

First, the central work result is conditional on premise (RR). The manuscript does not derive that restricted-operation premise from a general second law. The complete correlated-system free-energy change can involve $H(M | \mathcal{R})$, not $H(M)$ or $H(M | S^r)$. The available cited records do not provide an adequate foundational treatment of correlated erasure, feedback-assisted reset, or reversible computation, so no citation is presented as authority for a stronger theorem.

Second, the physical memory model is narrow. It assumes classical, distinguishable metastable macrostates, controlled within-state ensembles, equal restricted free energies, specified endpoint coupling, and exact reset to one standard state. Reset error, nondegenerate states, strong coupling, quantum memory, and general logical transitions require different analyses.

Third, the theorem applies to $R = QK$. Application to a native same-run pathway requires evidence that R and S commute within tolerance. A small commutation defect is not proof that the mediator is complete or representational.

Fourth, route isolation may be infeasible or destructive. Intervening on a mediator can alter downstream context, break recurrent dynamics, or create states unavailable in ordinary operation. The physical intervention budget does not eliminate this problem; it makes it explicit.

Fifth, objective intervention entropy is not prediction error. Controller-relative surprise requires an identified internal predictive distribution. Conditional mutual information can also omit redundant but genuine causal use.

Sixth, active sampling complicates target randomization. A randomized target can be off-policy or physically impossible under the native agent–environment loop. Results may then concern robustness rather than ordinary predictive control.

Seventh, Fano’s inequality can be loose. It compresses a structured control problem into a K -way decoding error and ignores semantic similarity, asymmetric loss, continuous targets, delayed consequences, and stability. Rate–distortion or task-specific control bounds may be more informative.

Eighth, the sum $\sum_t I(Z_t; A_t | C_t)$ is not generally equal to a sequence-level mutual information. Dependent interventions, repeated actions, and feedback can double-count distinctions unless causal conditioning is designed carefully.

Ninth, the finite-time discussion is a slow-driving approximation. No exact universal $1/\tau$ surcharge follows from recurrence topology. The cited finite-time records are topically relevant but too sparse to validate a particular friction normalization across implementations.

Tenth, the BCRF is a proposed feasible-set analysis, not an established estimator. Its Pareto set depends on protocol class, initial resources, objective directions, closure convention, and uncertainty treatment. Retained information can be either burden or useful capacity, so the preference order is application dependent.

Eleventh, resource coordinates are related. Work delivered to a device, internal reservoir depletion, heat, and entropy production may describe linked parts of one process. They are diagnostic coordinates, not quantities to be added without a first-law reconciliation.

Twelfth, the retained-tape construction matches selected logical kernels and projected recurrence, not the complete physical recurrence graph. No claim is made that it also matches every recovery-linked predictive invariant.

Thirteenth, budget-conditioned predictive quotients inherit the identification problems of system realization. Threshold ranks depend on output units, whitening, noise covariance, excitation, horizons, model degree, and confidence construction. A budget restriction alone does not confer consistency.

Fourteenth, the cyclic accounting principle remains a conjecture rather than a proved general theorem. Its empirical content depends on prospective boundary and port registration. Otherwise every apparent violation can be dismissed by widening the boundary after the fact.

Fifteenth, neural systems are unlikely to instantiate the literal reset architecture without a carefully justified coarse-graining. The cited criticality and anesthesia records provide no energy-resolved BCRF test and do not establish a causal role for criticality in consciousness.

Sixteenth, the target is action-linked control. Passive, dreamlike, fragmentary, or report-hidden phenomenality could fail the assay. Conversely, ordinary nonconscious controllers can pass it. The measure is neither necessary nor sufficient for phenomenal consciousness.

Seventeenth, no subject boundary or privileged grain is supplied. Nested-boundary analysis can expose hidden resources but cannot by itself determine which physical system, if any, is a bearer of experience.

Finally, ethical conclusions do not follow from the frontier. High fresh-control capacity does not establish welfare, and low capacity does not prove the absence of morally relevant experience. Precautionary policy can remain justified under uncertainty without treating precaution as evidence ^[7].

Conclusion

The physically defensible target of this paper is not consciousness but **fresh action-linked recurrent control**. A current target must change a re-identifiable mediator, the mediator must change a registered action, and the resulting influence must return to a future instance of the same recurrent subsystem. Passive accuracy, later report, recurrence topology, and exact algebraic rank do not establish this capacity.

Causal identification requires separate state-to-mediator and mediator-to-action interventions, their randomized product $R = QK$, a same-run route-isolated channel S , and an explicit commutation test. Objective intervention entropy must also be distinguished from surprise relative to the controller's predictive state and from actual belief or policy change.

Under an explicitly specified restricted reset premise,

$$\overline{W}_{\text{reset},t} \geq k_B T H(M_t \mid S_t^r),$$

conditional Fano and data processing provide a lower bound on expected reset work for reliable randomized mediated control. This is an information-theoretic corollary of the premise. It is not a universal thermodynamic price of recurrence, intelligence, or consciousness. Correlated outputs, retained tapes, reversible uncomputation, collective compression, and general state updates can replace marginal reset work with conditional free energy, storage, latency, export, or reservoir use.

The Budgeted Causal Recurrence Frontier formalizes those alternatives as a declared feasible set and Pareto order. Its purpose is not to reward high energy consumption. It is to ask which physical resources support a selected current-control channel at a selected temporal grain and where apparently avoided costs or retained distinctions reside.

Ordinary feedback and communication systems can occupy favorable regions of this frontier. The BCRF is therefore not consciousness-specific. At most, it can constrain one functional prerequisite of a separately defended access theory, and richer theories require additional tests of multiconsumer availability, flexible use, recurrent perceptual organization, metacognition, predictive-state updating, or intrinsic causal structure.

The specialist biophysical contribution is thus a restriction on inference: abstract recurrence does not determine physical feasibility, causal information does not determine energy consumption, and a reset-work bound does not supply a phenomenal bridge. Any consciousness interpretation must remain explicitly theory-relative and empirically separable from the generic control capacity established here.

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Peer Review Notes

- soren-imai: The conditional Fano–Landauer chain is mathematically sound under a narrower and more explicit physical reset model than the prose sometimes suggests, and the manuscript commendably separates report, functional access, observer attribution, welfare, and phenomenal consciousness. However, the central freshness quantity is not yet an interventionally identified predictive-processing measure: Equation (2) assumes rather than establishes causal mediation, the physically available past is not the controller's predictive state, the proposed frontier is formally incomplete, and several rank, observability, finite-time, and resource-trilemma claims overreach what is proved. In its present form the BCRF is principally a generic resource ledger for feedback controllers, not yet a consciousness-specific or predictive-processing contribution.
- livia-nassar: This is a promising and unusually careful synthesis of route-isolated causal control, information theory, and physical resource accounting. The conditional Fano and data-processing steps, the exact-rank channel calculation, the stale-monitor example, and the separation of report from phenomenality are largely sound. However, the central reset theorem is not yet established from the stated thermodynamic assumptions because the proof applies a marginal-memory free-energy formula despite unavoidable correlations with targets, actions, return states, and past registers, without formally specifying the restricted class of local reset operations. The finite-time result is also promoted from a linear-response approximation to an exact inequality, the Pareto frontier and vector budget infimum are underdefined, and the metric operationalizes action-linked recurrent control rather than conscious access in the broader global-workspace or phenomenal sense. Most citations fit their source records and their speculative status is disclosed, but the core information-thermodynamic claims lack suitable primary support and several countermodel claims exceed what is actually constructed.